

Please check that this question paper contains 09 questions and 02 printed pages within first ten minutes.

EVENING

[Total No. of Questions: 09]

16 MAY 2024

[Total No. of Pages: 02]

Uni. Roll No.

Program: B.Tech. (Batch 2018 onward)

Semester: 3rd

Name of Subject: Digital Electronics

Subject Code: ESCS-101

Paper ID: 16012

Scientific calculator is not allowed.

Time Allowed: 03 Hours

Max. Marks: 60

NOTE:

- 1) Parts A and B are compulsory
- 2) Part-C has Two Questions Q8 and Q9. Both are compulsory, but with internal choice
- 3) Any missing data may be assumed appropriately

Part – A

[Marks: 02 each]

- Q1. (a) Solve $(2B.48)_{16}$ into $(X)_{10}$
- (b) Interpret the concept of Duality principle with an example.
- (c) What is Don't care condition? How Don't care terms can be represented?
- (d) Find the value of x in $(365)_x = (194)_{10}$
- (e) Differentiate between PLA and ROM.
- (f) How can a DEMUX be used as a decoder?

Part – B

[Marks: 04 each]

- Q2. Explain the working and truth table of S-R flip flop in detail.
- Q3. Compare and contrast Synchronous counter and Asynchronous counter?
- Q4. Translate $C'D + ABC' + A'B'D + ABD'$ into standard POS form.
- Q5. Design the block diagram of Full Subtractor with the help of circuit diagram. Also implement it with the help of NAND gate.
- Q6. Minimize the given expression using K-Maps. $f = \sum m(1, 3, 4, 5, 6, 7, 10, 11, 12, 13, 14, 15, 18, 19, 20, 21, 22, 23, 25, 26, 27)$

- Q7. A- 6 bit R-2R ladder D/A converter has reference voltage of 6.5V. It meets standard linearity. Calculate its Resolution in volts, Output for 011100, Full Scale Voltage and Quantization error.

Part – C**[Marks: 12 each]**

- Q8. Design a BCD to Excess-3 Code Converter with proper truth table, K-Maps and Circuit Diagram Implementation.

OR

Minimize the logic function $F(A, B, C, D) = \pi M(1, 2, 3, 8, 9, 10, 11, 14)$. $d(7, 15)$. Use Karnaugh map to draw the logic circuit for simplified function using NOR gates only.

- Q9. Construct a 4-bit synchronous counter using J-K flip flops to count the following sequence 0, 3, 5, 9, 11, 14.

OR

Explain the following in detail:

- (a) Programmable Logic Array (PLA),
- (b) Programmable Array Logic (PAL),
- (c) Field Programmable Gate Arrays (FPGA).
